

Automated Evidence Synthesis Pipeline v2.0: Process Documentation

Overview

This pipeline automates the identification, ranking, and synthesis of academic evidence on nutrition interventions in low- and middle-income countries (LMICs). It combines multi-source programmatic literature search (PubMed + OpenAlex), structured scoring using authoritative metadata, and LLM-based review and synthesis.

Repository: github.com/anshaji/nutri-evidence-review (<https://github.com/anshaji/nutri-evidence-review>)

Stage 1: Literature Retrieval

Objective: Retrieve all meta-analyses, systematic reviews, and Cochrane reviews on nutrition interventions in LMICs, plus a parallel track for cost-effectiveness analyses and nutrition-sensitive interventions from the economics literature.

Sources:

Source	Role	API	Rate Limit
PubMed (E-Utilities)	Primary — biomedical literature	esearch + efetch	10 req/s (with API key)
OpenAlex	Supplementary — economics/development literature	REST API	0.3s between requests

Track A: Meta-Analyses & Systematic Reviews (PubMed)

Two-pass approach: - **Pass 1 (Primary tier):** Retrieves only confirmed meta-analyses using `meta-analysis[pt]` publication type filter - **Pass 2 (Supplementary tier):** Retrieves systematic reviews using `systematic review[pt]` filter

Query structure:

```
[intervention MeSH/keywords] AND [LMIC filter] AND [publication type filter]
```

LMIC filter (reusable across all queries):

```
("developing countries"[MeSH] OR "low income"[tiab] OR "middle income"[tiab] OR "LMIC"[tiab] OR "sub-saharan africa"[tiab] OR "south asia"[tiab] OR "southeast asia"[tiab])
```

Using "developing countries" [MeSH] is key — this is a curated MeSH descriptor that maps to all individual LMIC country names without needing to enumerate them.

The 12 intervention domain queries:

#	Domain	Key MeSH/Keywords
1	Micronutrient supplementation	"micronutrients" [MeSH] , "dietary supplements" [MeSH] , iron, zinc, vitamin A, folic acid
2	Food fortification	"food, fortified" [MeSH] , flour fortification, salt iodization, biofortification
3	Complementary feeding	"infant nutritional physiological phenomena" [MeSH] , complementary feeding, weaning
4	Breastfeeding promotion	"breast feeding" [MeSH] , "lactation" [MeSH] , kangaroo mother care
5	Acute malnutrition management	severe acute malnutrition, RUTF, CMAM
6	Maternal nutrition	"prenatal nutritional physiological phenomena" [MeSH] , maternal nutrition
7	WASH + nutrition	"water purification" [MeSH] , "hygiene" [MeSH] , "sanitation" [MeSH] AND nutrition/stunting
8	School feeding	school feeding, school meal, school nutrition
9	Growth monitoring & promotion	growth monitoring, "nutrition surveillance" [MeSH]
10	Deworming	"anthelmintics" [MeSH] , deworming AND nutrition/growth/anemia
11	Nutrition-sensitive agriculture	"agriculture" [MeSH] , homestead food production AND nutrition/diet
12	Integrated/multi-sectoral	"nutrition programs and policies" [MeSH] , nutrition-sensitive, nutrition-specific

Track B: Cost-Effectiveness Analyses (PubMed)

Separate track because CEAs are rarely meta-analyses — they come as original research articles.

```
("cost-benefit analysis"[MeSH] OR "cost-effectiveness"[tiab] OR "cost per DALY"[tiab]
OR "cost-benefit"[tiab] OR "cost effective"[tiab] OR "incremental cost"[tiab])
AND ("nutrition"[tiab] OR "malnutrition"[tiab] OR "stunting"[tiab]
OR "supplementation"[tiab] OR "fortification"[tiab]
OR "breastfeeding"[tiab] OR "complementary feeding"[tiab])
AND [LMIC filter]
```

No publication type restriction — CEAs are indexed as regular journal articles.

Track C: Supplementary Sweep (OpenAlex)

Retained for intervention domains PubMed covers poorly — the economics and development literature on nutrition-sensitive interventions:

#	Query Domain
1	Cash transfers + nutrition/food security/dietary diversity
2	Social protection + child nutrition/malnutrition
3	Food subsidies / public distribution systems
4	Conditional cash transfers + nutrition/child health

These queries use OpenAlex's full-text search with Boolean operators and target reviews/evidence syntheses.

Retrieval Parameters

Parameter	PubMed	OpenAlex
Max results per query	500 (via <code>retmax</code>)	500 (5 pages x 100)
Rate limiting	10/s with API key	0.3s between pages
Deduplication	PMID across queries	OpenAlex ID, then DOI-match against PubMed set
Abstract retrieval	<code>efetch</code> with <code>rettype=xml</code>	Inverted index reconstruction
Batch size	200 PMIDs per <code>efetch</code> request	100 per page

PubMed E-Utilities Method

1. **esearch** — submit query string, receive list of PMIDs + total count (JSON response)
2. **efetch** — submit PMIDs in batches of 200, receive full article records in XML
3. **Parse XML** — extract title, abstract (structured sections joined), MeSH terms, publication types, DOI, journal, year, authors
4. **Deduplicate** across all queries by PMID

Cross-Source Deduplication

After PubMed + OpenAlex retrieval: 1. Within PubMed: by PMID (when a paper appears in multiple queries, keep the highest-tier version) 2. Within OpenAlex: by OpenAlex work ID 3. Cross-source: by DOI (normalize: strip prefix, lowercase) - When DOI matches, **prefer PubMed record** (richer metadata: MeSH, publication types) - Merge `cited_by_count` and `is_open_access` from OpenAlex into the PubMed record

Output

A deduplicated candidate set with full metadata. Fields include: - `id`, `title`, `abstract`, `publication_year`, `journal`, `doi`, `pmid`, `openalex_id` - `source_db`: "pubmed" or "openalex" - `publication_type`: list from PubMed indexing (authoritative, not inferred) - `mesh_terms`: list of MeSH descriptor names - `study_type`: human-readable classification - `tier`: "primary" (meta-analysis), "supplementary" (systematic review), "cea", or "nutrition-sensitive" - `query_origin`: which of the 12+1+4 queries found it - `cited_by_count`: from OpenAlex cross-reference - `is_open_access`: boolean

Raw Response Archiving

All raw API responses are saved to `raw_responses/` for reproducibility: - PubMed: `pubmed_{tier}_{domain}_{timestamp}.xml` - OpenAlex: `openalex_{domain}_{timestamp}.json`

Stage 2: Relevance Scoring

Objective: Rank all collected papers by a composite relevance score that leverages authoritative metadata (MeSH terms, publication types) for PubMed papers and falls back to keyword heuristics for OpenAlex papers.

Method: Each paper receives a numerical score (0-85) computed as the sum of seven components:

Component 1: Study Design Authority (0-20 points)

For PubMed papers: Direct mapping from the authoritative `publication_type` field assigned by NLM indexers.

Publication Type	Points	Rationale
Meta-Analysis	20	Gold standard for evidence synthesis
Systematic Review	17	Structured evidence synthesis
Randomized Controlled Trial	14	High internal validity
Review	10	May include Cochrane/umbrella reviews
Practice Guideline	10	Authoritative clinical guidance
Clinical Trial	8	Experimental evidence
Comparative Study	6	Structured comparison
Evaluation Study	6	Program evaluation

When a paper has multiple publication types (e.g., both “Meta-Analysis” and “Systematic Review”), the **highest** score is taken.

For OpenAlex papers (keyword fallback, capped at 15): - "meta-analysis" → 15 - "umbrella review" → 15 - "cochrane" → 14 - "systematic review" → 12 - "cost-effectiveness" → 10 - "randomized controlled trial" → 8

Component 2: Topic Relevance (0-25 points)

For PubMed papers: Based on MeSH term set matching against curated intervention and outcome term sets.

Scoring logic: - Matches against intervention MeSH set (e.g., Micronutrients, Food Fortified, Breast Feeding): +4 per match, max 12 - Matches against outcome MeSH set (e.g., Nutritional Status, Growth Disorders, Malnutrition): +3 per match, max 9 - Bonus for having BOTH intervention AND outcome MeSH terms: +4 - Total capped at 25

For OpenAlex papers (keyword fallback, capped at 20): - Keyword presence matching: stunting (4), wasting (4), child nutrition (4), micronutrient (4), complementary feeding (4), malnutrition (3), breastfeeding (3), supplementation (3), fortification (3), food security (3)

Component 3: Setting Relevance (0-10 points)

For PubMed papers: MeSH geographic terms.

Relevant MeSH: “Developing Countries”, “Africa South of the Sahara”, “Asia, Southeastern”, “Asia, Southern”, “India”, “Bangladesh”, “Ethiopia”, “Nigeria”, etc.

Scoring: +4 per matching MeSH term, capped at 10.

For OpenAlex papers (keyword fallback): - "lmic" (5), "low-income" (4), "middle-income" (4), "developing countr" (4), "sub-saharan africa" (4), "south asia" (4) - Capped at 10

Component 4: Recency (0-10 points)

Step function based on publication year (unchanged from v1):

Paper Age	Points	Rationale
0-5 years	10	Current evidence
6-10 years	7	Still relevant
11-15 years	4	Foundational
16-20 years	2	Historical
20+ years	0	Likely superseded

Component 5: Citation Impact (0-12 points)

Step function based on `cited_by_count` (from OpenAlex cross-reference):

Citations	Points
500+	12
200-499	10
100-199	8
50-99	6
20-49	4
5-19	2
0-4	0

Component 6: Open Access Bonus (0-3 points)

Condition	Points
Open access	3

Component 7: Tier Bonus (0-5 points)

Condition	Points	Rationale
Paper from Track A Pass 1 (confirmed meta-analysis)	5	Prioritizes the purest evidence synthesis
All others	0	—

What Changed from v1 Scoring

Issue in v1	Fix in v2
Binary keyword matching for study type	Authoritative <code>publication_type</code> from PubMed NLM indexers
Keyword matching for topic relevance	MeSH term set matching (curated, hierarchical)
No negative signals	MeSH terms are assigned by human indexers — if “Developing Countries” isn’t tagged, the paper genuinely isn’t focused on LMICs
All study types mixed together	Two-tier system separates meta-analyses from systematic reviews at query time
Citation data only from OpenAlex	Cross-reference enrichment brings citation counts to PubMed papers
Additive keyword stacking	MeSH scoring uses caps per category to prevent gaming

Remaining Limitations

- **OpenAlex papers still use keyword fallback** — capped at lower scores but still noisy
- **Citation bias persists** for older papers (partially offset by recency score)
- **No full-text access** — scoring is based on metadata and abstracts only
- **MeSH indexing lag** — very recent papers may not yet have MeSH terms assigned
- **OpenAlex citation counts may lag** a few months behind reality

Stage 3: Database Export

Objective: Save scored papers in multiple formats for review and reuse.

Outputs:

File	Format	Contents
<code>papers_database.json</code>	JSON	All papers with full metadata, abstracts, and scores
<code>papers_ranked.csv</code>	CSV	All papers in tabular format, sorted by score descending

File	Format	Contents
top_papers_for_review.json	JSON	Top 100 papers (input for Stage 4)

Fields per paper:

Field	Type	Source
id	string	Canonical ID (pmid:XXXXX (pmid:XXXXX) or OpenAlex URL)
title	string	Paper title
abstract	string	Full abstract text
publication_year	int/null	Year of publication
journal	string	Journal name
doi	string/null	DOI (normalized)
pmid	string/null	PubMed ID
openalex_id	string/null	OpenAlex work ID
source_db	string	“pubmed” or “openalex”
study_type	string	Classified type (see below)
publication_type	list[string]	Raw from PubMed indexing
mesh_terms	list[string]	MeSH descriptor names
cited_by_count	int	From OpenAlex cross-reference
is_open_access	boolean	Open access status
query_origin	string	Which query found this paper
tier	string	“primary”, “supplementary”, “cea”, or “nutrition-sensitive”
relevance_score	float	Composite score (0-85)

Study type classification: - Systematic Review & Meta-Analysis - Meta-Analysis - Systematic Review - Cochrane Review - Umbrella Review - Cost-Effectiveness Analysis - RCT - Practice Guideline - Review - Article

Stage 3.5: Full-Text Retrieval (PMC Open Access)

Objective: Retrieve structured full-text content from PubMed Central for top-scored papers that have open-access PMC versions, enabling deeper LLM synthesis with effect sizes, subgroup analyses, and implementation details not present in abstracts.

Method:

1. **PMID to PMCID conversion** — Batch API call to NCBI ID Converter
(<https://www.ncbi.nlm.nih.gov/pmc/utis/idconv/v1.0/>) with up to 200 PMIDs per request

2. **PMC full-text fetch** — E-Utilities efetch with db=pmc for each PMCID
3. **XML parsing** — Extract structured sections (Introduction, Methods, Results, Discussion), tables (labels, captions, content), and concatenate into readable text
4. **Caching** — Raw XML saved to raw_responses/pmc/{PMCID}.xml for reproducibility and to avoid re-fetching

Coverage: - Typically 85/99 PubMed papers have a PMCID (PMC-indexed) - Of those, ~57 have downloadable full-text XML (body element present) - ~28 are publisher-restricted (Cochrane reviews and some Elsevier journals block XML download despite PMC indexing) - Final coverage: ~57% of top 100 papers have structured full text

Output fields added to papers: - fulltext_source: "pmc" or "abstract_only" - fulltext: dict with sections (list of {title, text}), tables (list of {label, caption, content}), and full_text (concatenated plain text) - pmcid: PMC identifier (if available)

Module: pipeline/fulltext_client.py

Known limitations: - NCBI ID Converter returns PMIDs as integers — must cast to str() for matching - Some publishers (Cochrane/Wiley, Elsevier) restrict PMC XML body access - No PDF parsing fallback (would require non-stdlib libraries)

Stage 4: LLM Review and Synthesis

Objective: Review the top-ranked papers and produce a structured list of nutrition interventions ranked by evidence strength, cost-effectiveness, and scalability through government-led programs in LMIC settings.

Input: The top 100 papers from Stage 3.5 (titles, abstracts, metadata, MeSH terms, tiers, scores, AND structured full text for ~57 papers).

Two synthesis modes:

Mode A: Abstract-Only Synthesis (lighter)

Uses only abstracts and metadata. Produces INTERVENTION_SYNTHESIS.md .

Mode B: Full-Text-Enhanced Synthesis (deeper)

Uses full text (Results sections, tables, subgroup analyses) where available, with abstracts as fallback. Produces FULL_INTERVENTION_SYNTHESIS.md .

LLM prompt (task framing):

You are reviewing the top 100 papers from a multi-source systematic search of the nutrition intervention literature in LMICs. Papers were retrieved from PubMed (using MeSH terms and publication type filters) and OpenAlex (for nutrition-sensitive literature). They are ranked by a composite score weighting study design authority, MeSH-based topic relevance, setting relevance, recency, citation impact, and tier assignment.

For 57 papers, you have access to structured full text (Results sections, tables, discussion) from PubMed Central. For the remaining 43 papers, you have abstracts and metadata only. Full-text papers are marked with `fulltext_source: "pmc"`.

Papers are organized into tiers: - **Primary tier:** Confirmed meta-analyses (highest evidence quality) - **Supplementary tier:** Systematic reviews - **CEA tier:** Cost-effectiveness analyses - **Nutrition-sensitive tier:** Cash transfers, social protection, food subsidies (from economics literature)

Your task:

1. **Identify distinct interventions.** Read across all papers and extract every discrete nutrition intervention. Group closely related variants under a single heading.
2. **Rate each intervention on three dimensions:**
 - **Evidence strength** (A/B/C): Based on number and quality of meta-analyses, consistency of findings, study type hierarchy
 - **Cost-effectiveness:** From CEA tier papers + general knowledge (Very High / High / Moderate / Low / Unknown)
 - **Scalability:** Evidence of national implementation, platform compatibility, infrastructure needs (Proven national / Proven subnational / Growing / Requires investment)
3. **Rank interventions into tiers:**
 - Tier 1: Strong evidence (A) + proven scalability + high cost-effectiveness
 - Tier 2: Strong evidence (A or B+) + scalable with investment
 - Tier 3: Promising evidence (B or C+) + plausible scaling pathway
4. **For each intervention, document:**
 - Evidence rating with justification (cite specific papers by PMID, year, journal, citations)
 - **Specific effect sizes with confidence intervals** (from full-text Results sections where available)
 - Mechanism of action
 - Government scaling pathway
 - Key supporting papers from the database

5. **Synthesize cross-cutting findings** (4-6 patterns across the evidence base)
6. **Produce a summary table** (Rank, Intervention, Evidence, Cost-Effectiveness, Scalability, Target)

Constraints: - Base synthesis on information in provided full text, abstracts, and metadata - For full-text papers: extract and report actual effect sizes (RR, OR, SMD, MD with 95% CIs) - For abstract-only papers: report only what is stated in the abstract - Do not fabricate effect sizes not present in the source material - Use MeSH terms and publication types to validate classification - Flag papers where full text was not available for key claims - Distinguish nutrition-specific from nutrition-sensitive interventions - Note where subgroup analyses (from full text) modify the headline finding

LLM review process: - Papers are read in batches (top 40, then 41-100) due to context window constraints - For each paper, the LLM reads: title, abstract, publication year, study type, publication types, MeSH terms, tier, citation count, journal, score - For full-text papers: additionally reads the Results section text and table data - MeSH terms help the LLM verify what intervention and population each paper actually studies - Tier assignment helps the LLM weight evidence appropriately (primary > supplementary) - Full-text access enables extraction of: pooled effect estimates, forest plot data, subgroup analyses, implementation context, and study heterogeneity metrics

Outputs: - INTERVENTION_SYNTHESIS.md — Abstract-only synthesis (lighter, 18 interventions) - FULL_INTERVENTION_SYNTH.md — Full-text-enhanced synthesis (deeper, 20 interventions with effect sizes)

Reproducibility

To rerun the pipeline:

```
cd nutri-evidence-review/
# Option A: use .env file (auto-loaded by config.py)
echo "NCBI_API_KEY=your_key_here" > .env
python3 fetch_papers.py

# Option B: export environment variable
export NCBI_API_KEY=your_key_here
python3 fetch_papers.py
```

Dependencies: Python 3.10+ standard library only (urllib, json, csv, xml.etree.ElementTree). No external packages required.

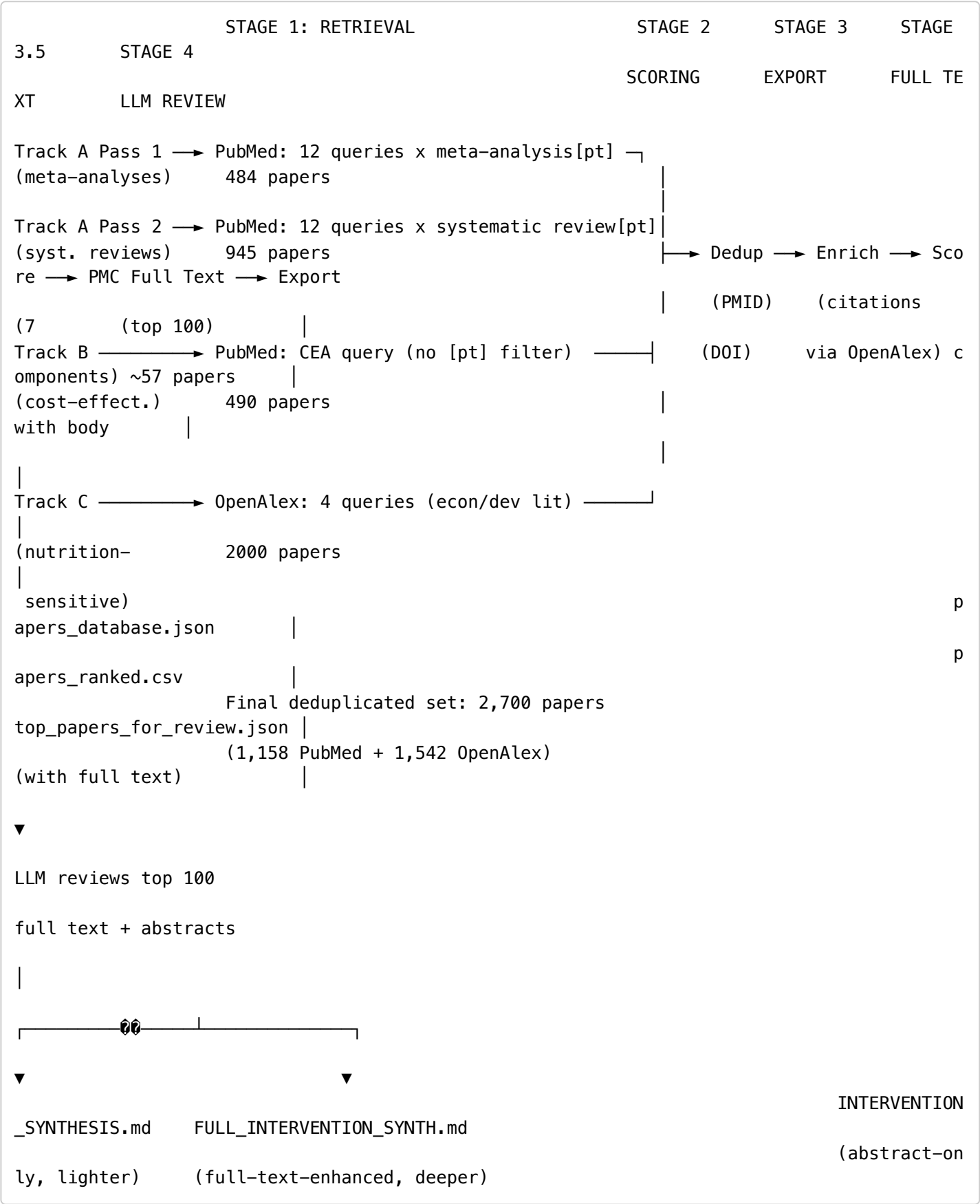
Runtime: ~3-4 minutes with NCBI API key (includes PMC full-text retrieval), ~7 minutes without key.

- Stage 1 (retrieval): ~50s PubMed + ~12s OpenAlex
- Stage 2 (dedup + enrich + score): ~35s
- Stage 3.5 (PMC full text): ~60-90s (cached on subsequent runs: instant)
- Export: ~5s

Caching: PMC full-text XML is cached in raw_responses/pmc/. Subsequent runs skip already-downloaded articles, making repeat runs much faster.

To modify: - **Search queries:** Edit `pipeline/queries.py` (`TRACK_A_QUERIES`, `TRACK_B_QUERY`, `TRACK_C_QUERIES`) - **Scoring weights:** Edit `pipeline/scoring.py` (`PUBTYPE_SCORES`, MeSH term sets, component functions) - **Number of papers for review:** Edit `TOP_N_FOR_REVIEW` in `pipeline/config.py` (default: 100) - **API configuration:** Edit `pipeline/config.py` or create `.env` file with `NCBI_API_KEY=your_key` - **Full-text retrieval:** Edit `pipeline/fulltext_client.py` (batch sizes, parsing logic)

Pipeline Architecture Diagram



File Structure

```
nutri-evidence-review/
├── fetch_papers.py           # Entry point: python3 fetch_papers.py
├── pipeline/
│   ├── __init__.py
│   ├── config.py           # API keys, .env loader, rate limits, output paths
│   ├── queries.py          # All query definitions (12 + 1 + 4)
│   ├── models.py           # Paper TypedDict (unified schema)
│   ├── pubmed_client.py    # E-Utilities: esearch, efetch, XML parsing
│   ├── openalex_client.py  # Track C: OpenAlex fetcher
│   ├── citation_enrichment.py # DOI-based cross-reference for citation counts
│   ├── dedup.py            # 3-phase deduplication logic
│   ├── scoring.py          # 7-component scoring algorithm
│   ├── fulltext_client.py  # PMC full-text retrieval (Stage 3.5)
│   └── main.py             # Orchestrator: runs all stages
├── .env                     # API key (gitignored)
├── raw_responses/          # Saved API responses (gitignored)
│   ├── pubmed_*.xml        # PubMed efetch responses
│   ├── openalex_*.json     # OpenAlex API responses
│   └── pmc/                # PMC full-text XML (one file per PMCID)
│       ├── PMC6572871.xml
│       └── ...
├── papers_database.json    # Full output (gitignored)
├── papers_ranked.csv       # CSV output (gitignored)
├── top_papers_for_review.json # Top 100 with full text (gitignored)
├── PROCESS_DOCUMENTATION.md # This file
├── INTERVENTION_SYNTHESIS.md # Abstract-only synthesis (Mode A)
└── FULL_INTERVENTION_SYNTH.md # Full-text-enhanced synthesis (Mode B)
```

Version History

Version	Date	Changes
v1.0	2026-05-22	Initial pipeline: OpenAlex only, 16 keyword queries, binary keyword scoring
v2.0	2026-05-24	Multi-source (PubMed + OpenAlex), MeSH-based scoring, two-tier MA/SR separation, citation enrichment, modular architecture
v2.1	2026-05-24	Added Stage 3.5: PMC full-text retrieval for top papers. Full-text-enhanced synthesis (FULL_INTERVENTION_SYNTH.md). .env auto-loading. API key masked from source.